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Comparison of the Effects of Velocity-Based Training Methods and Traditional 1RM-Percent-Based Training Prescription on Acute Kinetic and Kinematic Variables

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Purpose: To compare kinetic and kinematic data from 3 different velocity-based training sessions and a 1-repetition-maximum (1RM)-percent-based training (PBT) session using full-depth, free-weight back squats with maximal concentric effort. ***Methods:*** Fifteen strength-trained men performed 4 randomized resistance-training sessions 96 h apart: PBT session involved 5 sets of 5 repetitions using 80% 1RM; load–velocity profile (LVP) session contained 5 sets of 5 repetitions with a load that could be adjusted to achieve a target velocity established from an individualized LVP equation at 80% 1RM; fixed sets 20% velocity loss threshold (FS_{VL20}) session consisted of 5 sets at 80% 1RM, but sets were terminated once the mean velocity (MV) dropped below 20% of the threshold velocity or when 5 repetitions were completed per set; and variable sets 20% velocity loss threshold session comprised 25 repetitions in total, but participants performed as many repetitions in a set as possible until the 20% velocity loss threshold was exceeded. ***Results:*** When averaged across all repetitions, MV and peak velocity (PV) were significantly ($P < .05$) faster during the LVP (MV effect size [ES] = 1.05; PV ES = 1.12) and FS_{VL20} (MV ES = 0.81; PV ES = 0.98) sessions compared with PBT. Mean time under tension (TUT) and concentric TUT were significantly less during the LVP sessions compared with PBT. The FS_{VL20} sessions had significantly less repetitions, total TUT, and concentric TUT than PBT. No significant differences were found for all other measurements between any of the sessions. ***Conclusions:*** Velocity-based training permits faster velocities and avoids additional unnecessary mechanical stress but maintains similar measures of force and power output compared with strength-oriented PBT in a single training session.

Keywords: load–velocity relationship, back squat, load monitoring, training volume, resistance training

Determining training loads within a periodized program allows strength coaches to target specific attributes, optimize adaptation, and allow for recovery.[1] A common method to determine resistance-training loads, known as percent-based training (PBT), prescribes relative submaximal loads from a 1-repetition maximum (1RM). Even though 1RMs are valid, reliable, and require no monitoring equipment, they are time consuming when conducted with large groups. Moreover, maximal strength can fluctuate daily when an individual is fatigued or significantly increase within a few weeks due to training adaptation.[2] Consequently, training when fatigued or continued training based on an outdated 1RM may not optimize the neuromuscular stimuli required to maximize adaptation. For these reasons, alternative methods for prescribing training loads have been established.

For example, repetition maximum (RM) training requires an athlete to perform a prescribed number of repetitions in a set (eg, 8–10 RM) until concentric muscular failure (RM training). Although this method accounts for sessional adjustment of training load, it may require an athlete to perform multiple sets to reach the target RM load. Furthermore, research suggests that RM training sets can reduce the force-generating capacity in subsequent training sets and may diminish maximal strength development in well-trained athletes.[3–5] More recently, adjusting training loads based on an athlete's rating of perceived exertion has become an alternative to PBT, because it allows for the modification of sessional loads based on an athlete's perceived readiness to train.[6,7] Although rating of perceived exertion-based methods are valid and reliable, they can be problematic because they are subjective and also require a prescribed number of repetitions in a set to be completed until adjustments can be made. Therefore, an approach that uses instantaneous repetition feedback to objectively prescribe training loads could optimize adaptation and avoid training to failure.

Due to advancements in commercially available kinetic and kinematic measuring devices, it is now possible to provide instantaneous feedback during training for numerous variables, such as movement velocity. Accordingly, recent literature has explored the use of immediate feedback employing velocity-based training (VBT) methods to objectively manipulate resistance-training loads within a training session.[8–11] If an exercise is performed with maximal concentric effort and fatigue ensues, the barbell velocity of the movement will decrease within a set.[3,5] Because greater movement velocities with a given load increase the neuromuscular stimuli and adaptations to strength training,[8] decreases in movement velocity can be detrimental. Therefore, VBT can be used to monitor barbell velocity and avoid performing repetitions to concentric muscular failure if the training target is to optimize maximal strength and power development, particularly in well-trained athletes.[4]

One VBT approach is characterized by the cessation of a working set if mean velocity (MV) of a repetition falls below a predetermined velocity loss threshold.[12] For example, Padulo et al[2] implemented a 20% velocity loss threshold and showed

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that maintaining at least 80% of MV during training results in greater increases in bench press 1RM compared with RM training. Similarly, Pareja-Blanco et al[9] also showed that using a 20% velocity loss threshold resulted in similar increases in strength but greater increases in power output compared with a 40% velocity loss threshold. To further elaborate, 2 variations of the velocity loss threshold method have been introduced.[12] The variable sets velocity loss threshold (VS_{VL}) method includes a fixed training load and total number of repetitions, but allows for an indefinite number of sets, each finishing when a repetition drops below a predetermined maximum percent velocity loss.[11] This method allows an athlete to spread the total number of repetitions across multiple sets, allowing for greater recovery time, enabling an athlete to perform all of the prescribed repetitions with faster movement velocities. Comparatively, the fixed sets velocity loss threshold (FS_{VL}) method includes a predetermined training load and number of sets, but does not have a prescribed number of repetitions. This method requires an athlete to perform repetitions in a set until they are no longer able to produce the required velocity.

Importantly, MV and individualized load–velocity profiles (LVPs) have been shown to be reliable,[13–15] yet no research has explored the use of the LVP as a method of adjusting training load. It is proposed that if velocity targets are not met according to individualized LVP during a training session, then training load can be adjusted to meet these targets.[14] For example, if velocity is lower than the velocity from the individualized LVP, the training load can be reduced. Alternatively, if the velocity output during a training session is faster than the target velocity, the training load can be increased. Previous VBT research has individualized training volume prescription (number of repetitions per set),[9–11] but, notably, no research has used velocity to individualize training load prescription (load–velocity relationship). Additionally, participants within these studies have used a Smith machine and not a large mass free-weight barbell exercise. This is an important distinction, because free-weight exercises are extensively utilized in practice with most athletes and often require movements in both the vertical and horizontal planes, which may produce different velocity, force, and power outputs. Furthermore, no studies have compared VBT to more traditional PBT methods.

Therefore, the objective of this study was to compare the effects of the LVP, FS_{VL}, VS_{VL}, and PBT methods on the mechanical capacities of the lower body during a typical strength-oriented training session in a free-weight exercise. Based on the results of previous research,[9–11] we hypothesized that all 3 VBT methods would result in greater movement velocities and power outputs, but the LVP and FS_{VL} methods would result in the completion of less work and time under tension compared with a PBT session, because it was conceivable that lighter loads (LVP method) or fewer repetitions (FS_{VL} method) would be completed.

Methods

Participants

Fifteen resistance-trained male volunteers participated in this study (age: 25.1 [3.9] y, height: 179.7 [6.7] cm, and body mass: 83.9 [10.6] kg) and performed the full-depth back squat with a mean knee flexion angle of 121.8° (9.4°) as measured by a handheld plastic goniometer. Their mean 1RM was 151.3 (22.2) kg, which was normalized to 1.83 (0.29)/kg of body mass (range = 1.55–2.43/kg of body mass). The participants had 6.7 (2.2) years of resistance-training experience ranging from 1 to 3 sessions per week and were free from musculoskeletal injuries. Each participant gave written informed consent prior to volunteering, which was in accordance with the Edith Cowan University human research ethics board. The protocols for this study also adhered to the Code of Ethics of the World Medical Association (Declaration of Helsinki).

Experimental Design

A randomized crossover design was chosen in which volunteers completed all 4 conditions. Participants visited the laboratory for a 1RM session, LVP assessment, and 4 randomized strength-oriented training sessions. Participants were afforded a 48-hour rest following the 1RM and LVP assessments but a 96-hour rest between the 4 testing sessions to allow for sufficient recovery. Experimental sessions included: 1RM PBT; LVP; fixed sets 20% velocity loss threshold (FS_{VL20}); variable sets 20% velocity loss threshold (VS_{VL20}) (Table 1).

Session 1: 1RM Assessment

Participants performed all repetitions in a power cage (Fitness Technology, Adelaide, Australia) using a 20-kg barbell (Eleiko®, Halmstad, Sweden). Prior to the 1RM assessment, participants performed warm-up procedures consisting of 5 minutes pedaling on a cycle ergometer (Monark 828E cycle ergometer; Monark Exercise, Vansbro, Sweden) at 60 rpm and 60 W, 3 minutes of dynamic stretching, and 10 full-depth body-weight squats. Participants then commenced the 1RM assessment, comprising sets estimated at 20% 1RM (3 repetitions), 40% 1RM (3 repetitions), 60% 1RM (3 repetitions), 80% 1RM (1 repetition), and 90% 1RM (1 repetition).[16] This was followed by the first 1RM attempt, with five 1RM attempts permitted. After successful 1RM attempts, barbell load was increased in consultation with the participant between 0.5 and 2.5 kg. The last successful lift with correct technique and full depth was classified as the 1RM load. A two-minute passive rest was allocated between all warm-up sets, but 3 minutes were allocated between 1RM attempts. Participants were

Table 1 Description of the Experimental Sessions

Group	Load (%1RM)	Sets	Repetitions	Total repetitions	Set completion
PBT	80%	5	5	25	Forced completion
LVP	80%; adjust load by 5% 1RM if set is ±0.06 $m \cdot s^{-1}$ LVP at 80%	5	5	25	Forced completion
FS_{VL20}	80%	5	—	—	20% velocity decline
VS_{VL20}	80%	—	—	25	20% velocity decline

Abbreviations: 1RM, 1-repetition maximum; FS_{VL20}, fixed sets 20% velocity loss threshold; LVP, load–velocity profile; PBT, traditional 1RM-percent-based training; VS_{VL20}, variable sets 20% velocity loss threshold.

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required to keep the barbell in constant contact with the superior aspect of the trapezius muscle and their feet in contact with the ground at all times. The eccentric phase of the squat was performed in a controlled manner, but once full knee flexion was achieved, the concentric phase was completed as quickly as possible.

Session 2: LVP Assessment

Recent research has reported that MV of the free-weight back squat is reliable at 20% 1RM, 40% 1RM, 60% 1RM, 80% 1RM, and 90% 1RM but not at 100% 1RM.[14] Therefore, it was recommended that individualized LVPs should be developed using MV from 20% 1RM to 90% 1RM. Consequently, participants came to the laboratory and performed the same cycling and dynamic warm-up protocols as in session 1, followed by back squat sets using 20% 1RM (3 repetitions), 40% 1RM (3 repetitions), 60% 1RM (3 repetitions), 80% 1RM (1 repetition), and 90% 1RM (1-repetition) with 2-minute passive recovery given between sets. The 1RM assessment (session 1) allowed for accurate relative 1RM loads to be lifted for the LVP assessment. For sets that included more than 1 repetition (ie, 20% 1RM, 40% 1RM, and 60% 1RM), the repetition with the fastest MV was utilized for the LVP.[17] The individualized LVPs were constructed by plotting MV against relative load and then applying a line of best fit to the data (Microsoft Excel 2016; Microsoft, Redmond, WA). A linear regression equation was then calculated and used to modify the training load in the LVP experimental session.

Sessions 3 to 6: Experimental Sessions

At the commencement of the randomized sessions, participants performed the identical cycling, dynamic stretching, and bodyweight squat warm-up protocols as in sessions 1 and 2. Participants then performed warm-up squat sets with maximal concentric effort using 20% 1RM (3 repetitions), 40% 1RM (3 repetitions), 60% 1RM (3 repetitions), and 80% 1RM (1 repetition) prior to all experimental testing sessions. Following these squat sets, a 2-minute rest was given before the commencement of an experimental session. The rest period between each training set was 2 minutes, and the time between repetitions within a set was approximately 2 seconds for both the VBT sessions and the PBT session. All squat repetitions in every session were performed with a self-selected, controlled eccentric velocity to full depth (knee angle = 122.5° [8.3°], squat depth = 707 [57] mm), which did not differ between sets or sessions ($P > .05$). The concentric phase for every repetition in all sessions was performed with maximal effort immediately after the eccentric phase. In addition, all participants were verbally encouraged to perform each repetition with maximal effort but no participant was provided or able to observe any velocity feedback in any session. However, participants were instructed to terminate a set if velocity targets were not met (FS_{VL20} or VS_{VL20} session), and the load was adjusted if velocity targets were not met (LVP session), indicating that the subjects could figure out whether the velocity of their previous set was above or below the threshold.

The PBT session involved 5 sets of 5 repetitions (25 repetitions) using 80% 1RM. During this protocol, MV was measured, but did not dictate changes in the external load, number of sets, or number of repetitions.

For the LVP training session, participants performed 5 sets of 5 repetitions, but with a load that (when performed with maximal intent) achieved an individualized target velocity that was established from the individual's LVP regression equation at 80% 1RM (established in session 2). If the difference in MV during the warm-up at 80% 1RM was ±0.06 $m \cdot s^{-1}$ compared with the target velocity at 80% 1RM according to the individualized LVP (as reported by Banyard et al[14]), the first set's training load was adjusted by ±5% 1RM. Otherwise, the relative load for the initial training set was kept at 80% 1RM. Once a set of 5 repetitions was completed, the load for the subsequent set could be adjusted by ±5% 1RM if the average of the MV for the 5 repetitions of the previous set was ±0.06 $m \cdot s^{-1}$outside of the target velocity at 80% 1RM according to the LVP. In this manner, all participants completed 25 repetitions, but the load of subsequent sets was adjusted according to the MV of the preceding set's repetitions.

The FS_{VL20} session was similar to the PBT session in which individuals completed 5 sets using 80% 1RM. However, sets were terminated once the MV of a repetition dropped below 20% of the threshold velocity or when 5 repetitions in the set were completed at or above the threshold velocity. When a set was terminated, the participant was instructed to cease squatting and rack the barbell. In this manner, it was possible for participants to either complete all 25 repetitions or fewer repetitions.

In the VS_{VL20} session, participants completed 25 repetitions in total, but they performed as many repetitions in a row as possible until the threshold velocity loss (20%) was exceeded. In this manner, participants completed 25 repetitions in as few sets as possible. The 20% velocity loss threshold for the FS_{VL20} and VS_{VL20} sessions was determined from the MV of the single repetition performed at 80% 1RM in the experimental protocol warm-up.

Data Acquisition

All kinetic and kinematic data were collected during the concentric phase of the squat unless noted otherwise using methodology similar to previous research.[18–20] This included MV and peak velocity (PV) measures that were captured from 4 linear position transducers (PT5A-250; Celesco Transducer Products, Inc, Chatsworth, CA) mounted to the top of the power cage, with 2 positioned anterior and above the collar of the barbell and 2 positioned posterior and above the collar of the barbell. The eccentric phase of each repetition commenced at zero displacement (standing) and was completed at maximal displacement (greatest descent), whereas the concentric phase began at maximal displacement and terminated at zero displacement. Time under tension (TUT) was calculated by adding the time spent during the eccentric (ETUT) and concentric (CTUT) phases of each repetition. The sum of the TUT for the respective phases was also calculated for the session (sETUT, sCTUT, and sTUT; Figure 1). Mean force (MF) and peak force (PF) were acquired from the quantification of ground reaction forces with the use of a force plate (BP6001200; AMTI, Watertown, MA). Mean power (MP) was calculated as an average, and peak power (PP) measures were calculated from the product of force and bar velocity. Mean total work (MW) and total session work (TW) were calculated by integrating the area under the force–displacement curve during the eccentric and concentric phases of each repetition.[21] The total session load (TL) and mean session load (ML) were also established. The linear position transducer and force plate data were collected through a BNC-2090 interface box with an analog-to-digital card (NI-6014; National Instruments, Austin, TX) and sampled at 1000 Hz. All data were collected and analyzed using a customized LabVIEW program (version 14.0; National Instruments). All signals were filtered with a fourth-order low-pass Butterworth filter with a cutoff frequency of 50 Hz. The total tension on the barbell as a result

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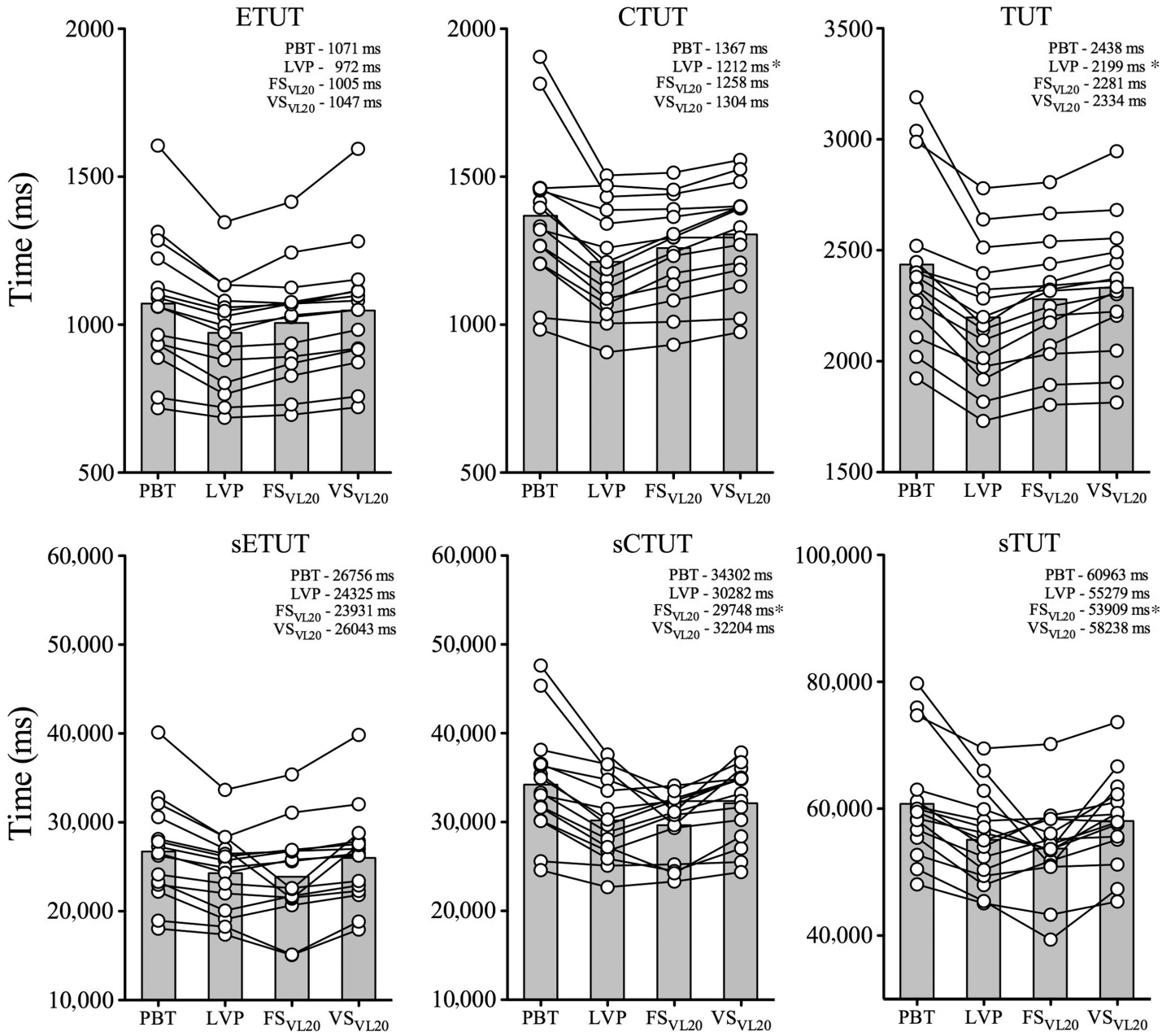

Figure 1 — Individual variation during the 1RM PBT, LVP, FS_{VL20}, and VS_{VL20} sessions for values of mean ETUT, mean CTUT, mean TUT, total sETUT, total sCTUT, and total sTUT. The shaded bars indicate the group mean, and the figure legend contains the numerical mean group values for the experimental sessions. 1RM indicates 1-repetition maximum; CTUT, concentric time under tension; ETUT, eccentric time under tension; FS_{VL20}, fixed sets 20% velocity loss threshold; LVP, load–velocity profile; PBT, percent-based training; sCTUT, session CTUT; sETUT, session ETUT; sTUT, session TUT; TUT, total time under tension; VS_{VL20}, variable sets 20% velocity loss threshold. *Significant difference between an experimental session and PBT.

of the transducer attachments (23.0 N) was accounted for in all calculations. Mean values of velocity, force, and power were determined as the average of the data collected during the concentric phase of the repetition. By contrast, peak values of velocity, force, and power were determined as the maximum value during the concentric phase of the lift. MV, PV, MF, PF, MP, PP, ML, MW, ETUT, CTUT, and TUT were calculated as the average of all repetitions for each individual within a session (Figures 1–5), whereas TL, TW, sETUT, sCTUT, and sTUT were calculated by totaling the respective data from all repetitions for each individual in a session (Figures 1 and 5).

Statistical Analyses

For all dependent variables, a repeated-measures analysis of variance was used to identify any differences between the 4 experimental protocols, with a type I error rate set at $\alpha \leq .05$. Holm's sequential Bonferroni post hoc comparisons were used when appropriate.[22] Effect sizes (ESs; ±95% confidence intervals) were calculated using Cohen *d*, which was interpreted as trivial (≤0.2), small (0.21–0.59), moderate (0.60–1.19), large (1.2–1.99), or very large (≥2.0).[23] Data analyses were performed using a statistical software package (SPSS version 22.0; IBM, Armonk, NY). Data are reported as mean (SD) unless stated otherwise.

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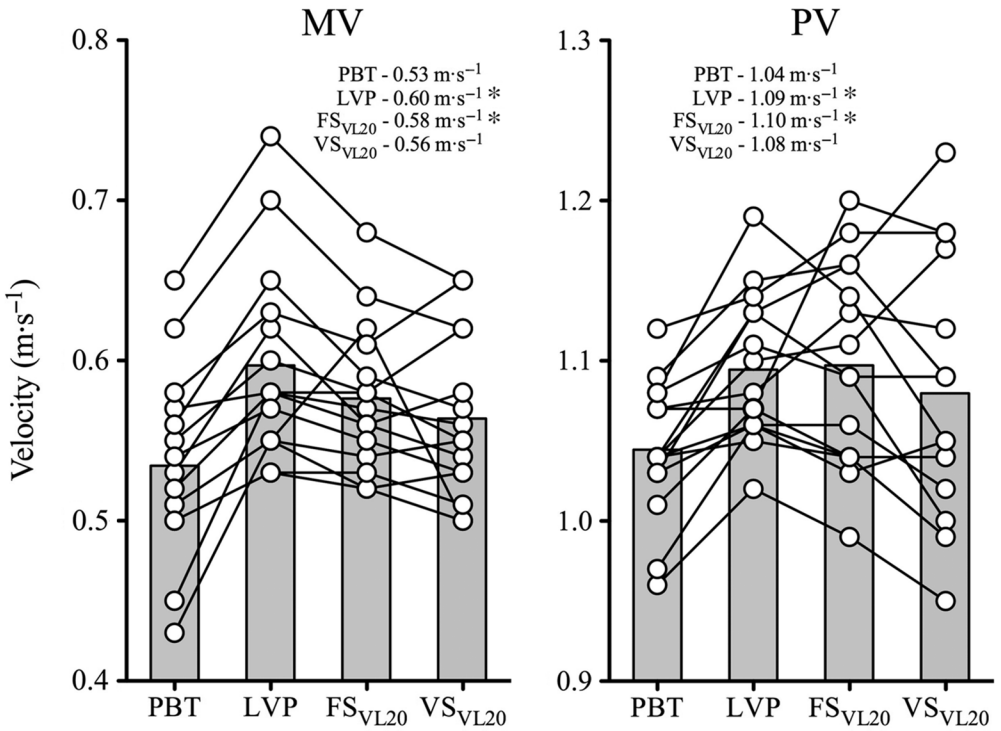

Figure 2 — Individual variation of mean repetition values of MV and PV for the 1RM PBT, LVP, FS_{VL20}, and VS_{VL20} sessions. The shaded bars indicate the group mean, and the figure legend contains the numerical group velocities for the experimental sessions. 1RM indicates 1-repetition maximum; FS_{VL20}, fixed sets 20% velocity loss threshold; LVP, load–velocity profile; MV, mean velocity; PBT, percent-based training; PV, peak velocity; VS_{VL20}, variable sets 20% velocity loss threshold. *Significant difference between an experimental session and PBT.

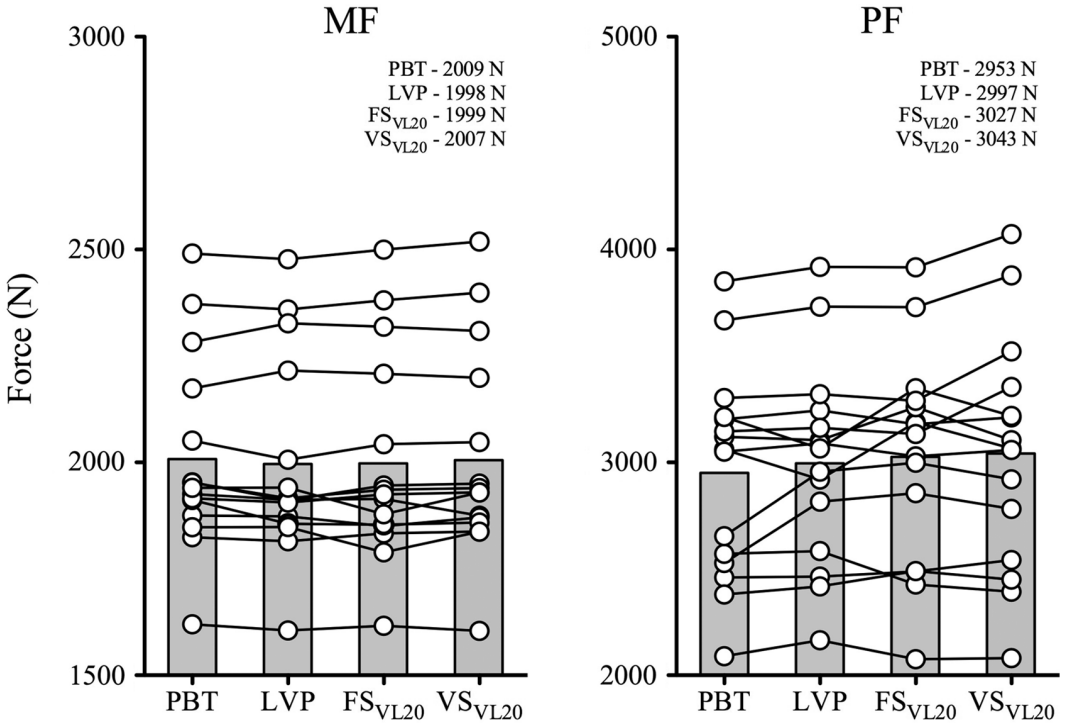

Figure 3 — Individual variation of mean repetition values of MF and PF for the 1RM PBT, LVP, FS_{VL20}, and VS_{VL20} sessions. The shaded bars indicate the group mean, and the figure legend contains the numerical group force values for the experimental sessions. 1RM indicates 1-repetition maximum; FS_{VL20}, fixed sets 20% velocity loss threshold; LVP, load–velocity profile; MF, mean force; PBT, percent-based training; PF, peak force; VS_{VL20}, variable sets 20% velocity loss threshold.

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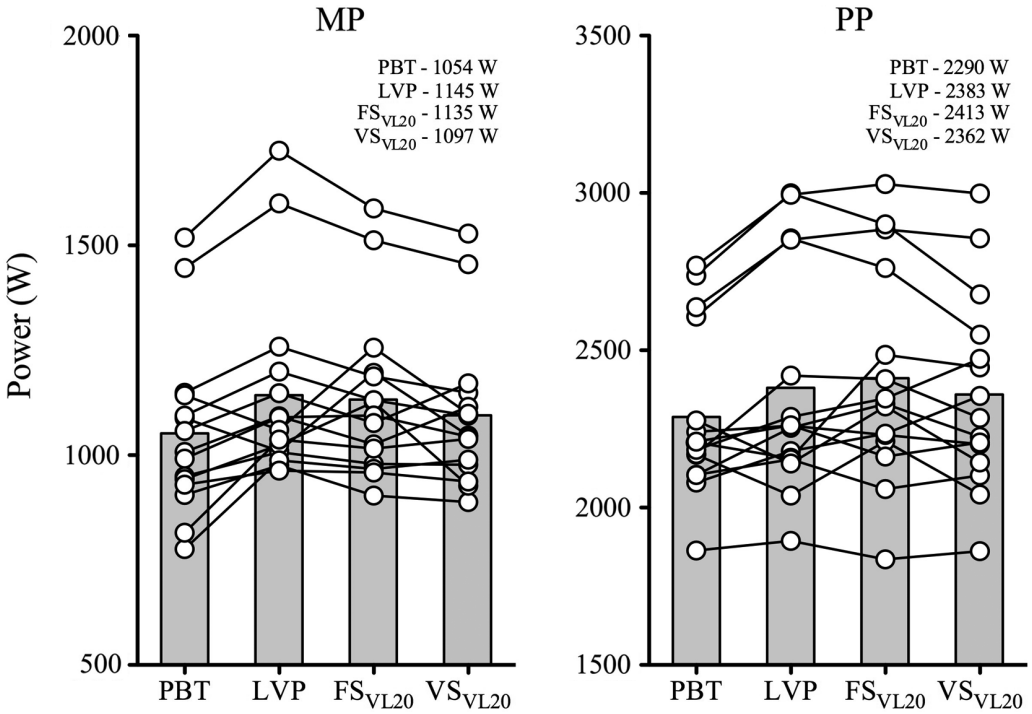

Figure 4 — Individual variation of mean repetition values of MP and PP for the 1RM PBT, LVP, FS_{VL20}, and VS_{VL20} sessions. The shaded bars indicate the group mean, and the figure legend contains the numerical group power values for the experimental sessions. 1RM indicates 1-repetition maximum; FS_{VL20}, fixed sets 20% velocity loss threshold; LVP, load–velocity profile; MP, mean power; PBT, percent-based training; PP, peak power; VS_{VL20}, variable sets 20% velocity loss threshold.

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Results

The load, number of sets, number of repetitions per set, and total session repetitions for each experimental session are shown in Table 2. There was no adjustment of training load for the first set of the LVP session for any participants, indicating the MV at 80% 1RM in the warm-up was within the smallest detectable difference range (±0.06 $m \cdot s^{-1}$ at 80% 1RM). Significantly fewer repetitions were performed during FS_{VL20} (23.6 [2.0] repetitions) compared with all other sessions (25 repetitions) (Table 2). During VS_{VL20}, participants completed the 25 repetitions in 4.3 (0.9) sets (range = 3–6 sets; Table 2). Session time was significantly shorter during VS_{VL20} (9:02 [1:55] min) than PBT (10:36 [0:19] min), LVP (10:34 [0:22] min), and FS_{VL20} (10:21 [0:55] min). Compared with PBT (MV: 0.53 [0.06] $m \cdot s^{-1}$; PV: 1.04 [0.04] $m \cdot s^{-1}$), MV and PV were significantly faster during LVP (MV: 0.60 [0.06] $m \cdot s^{-1}$, ES = 1.05; PV: 1.09 [0.04] $m \cdot s^{-1}$, ES = 1.12) and FS_{VL20} (MV: 0.58 [0.05] $m \cdot s^{-1}$, ES = 0.81; PV: 1.10 [0.06] $m \cdot s^{-1}$, ES = 0.98) (Figure 2). TUT and CTUT were significantly less during LVP compared with PBT (Figure 1). Significant differences were also observed between PBT and FS_{VL20} for sTUT and sCTUT (Figure 1). There were no significant differences between any of the sessions for all other variables.

Discussion

The major findings from the present study were that participants could sustain significantly faster MV and PV for repetitions performed during LVP and FS_{VL20} compared with PBT. In addition, the same 2 VBT methods allowed participants to perform repetitions with significantly less mechanical stress (CTUT, TUT, sCTUT, and sTUT) while still completing similar amounts of work (MW and TW) to PBT. The FS_{VL20} session also resulted in significantly fewer repetitions than all other methods. Furthermore, no significant differences were observed for measurements of force (MF and PF), power (MP and PP) and training load lifted (ML and TL) between any of the experimental sessions. Consequently, the LVP and FS_{VL20} methods appear to be more favorable than PBT for athletes performing strength-oriented training sessions due to faster movement velocities, less mechanical stress, but similar measures of force, power, work, and training load.

The significantly higher MV (ES = 1.05) and PV (ES = 1.12) observed during the LVP session compared with the PBT session can be attributed to subtle decreases in load (~5% 1RM) between sets, yet the total load lifted was not significantly less. By comparison, the significantly higher movement velocity observed during FS_{VL20} compared with PBT (MV: ES = 0.81; PV: ES = 0.98) were due to the strict 20% velocity loss threshold, which resulted in the completion of significantly fewer repetitions (23.6 [2.0] vs 25). It is difficult to determine the optimal resistance-training dose for maximizing strength and power development because there are so many factors that influence adaptation. However, research investigating this phenomenon in the back squat and weight lifting exercises has established that performing a moderate volume of repetitions could be more beneficial than performing an unnecessarily high number of repetitions (high volume).[24,25] For example, Pareja-Blanco et al[10] had 16 resistance-trained professional male soccer players perform 6 weeks (18 sessions, ranging from ~50 to ~70% 1RM) of back squat training. The players were evenly assigned into 2 groups, which differed by a 15% or 30% velocity loss threshold in each training set. Subsequently, the 15% velocity loss group trained with significantly fewer repetitions (total repetitions: 251.2 [55.4] vs 414.6 [124.9];

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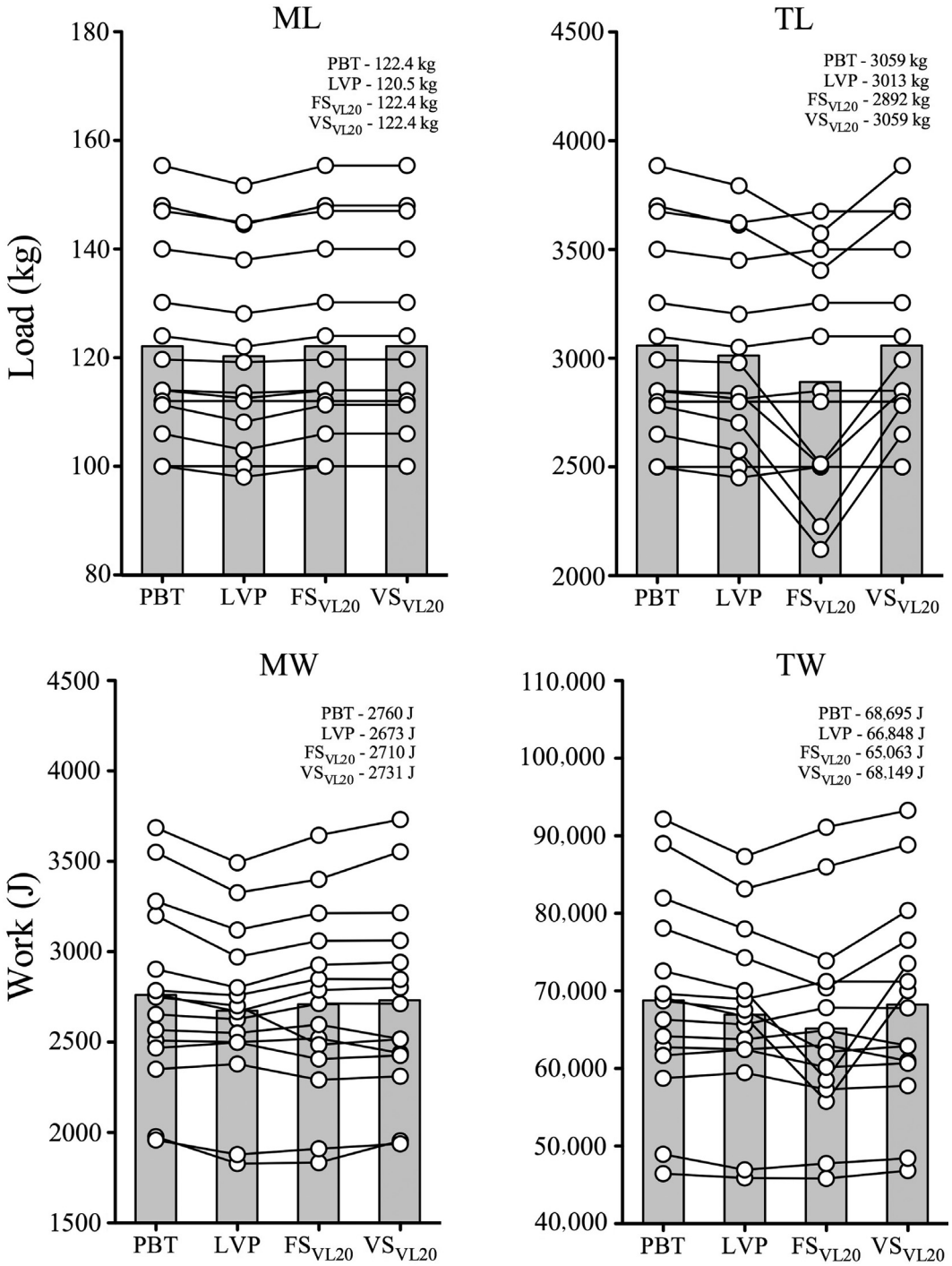

Figure 5 — Individual variation during the 1RM PBT, LVP, FS_{VL20}, and VS_{VL20} sessions for values of ML, MW, TL, and TW. The shaded bars indicate the group mean, and the figure legend contains the numerical mean group values for the experimental sessions. 1RM indicates 1-repetition maximum; FS_{VL20}, fixed sets 20% velocity loss threshold; LVP, load–velocity profile; ML, mean repetition load; MW, mean repetition work; PBT, percent-based training; TL, total session load; TW, total session work; VS_{VL20}, variable sets 20% velocity loss threshold.

mean repetitions/set: 6.0 [0.9] vs 10.5 [1.9]) and at faster movement velocities (MV: 0.91 [0.01] $m \cdot s^{-1}$ vs 0.84 [0.02] $m \cdot s^{-1}$), yet had significantly greater increases in maximal strength (estimated 1RM squat) and power output (countermovement-jump height) compared with the 30% velocity loss group. In light of these findings, coaches can monitor velocity and employ velocity loss thresholds so that immediate feedback can help inform accurate training volume decisions where limiting repetitions performed at slower movement velocities and maximizing the number of repetitions performed at faster velocities may produce greater increases in strength and power adaptations over time.

The LVP and FS_{VL20} sessions also resulted in significantly less mechanical stress compared with the PBT session, evidenced by less CTUT (Figure 1). In accordance with the load–velocity relationship, the lower mechanical stress observed during the LVP session was a consequence of the subtle training load

Table 2 Mean (SD) Description of Each Experimental Session

Session	Load, kg	Sets	Repetitions per set						Total repetitions
			1	2	3	4	5	6	
PBT	122.4 (17.8)	5	5	5	5	5	5	—	25
LVP	120.5 (17.2)	5	5	5	5	5	5	—	25
FS_{VL20}	122.4 (17.8)	5	5	5	4.8 (0.4)	4.5 (0.7)	4.2 (0.9)	—	23.6 (2.0)
VS_{VL20}	122.4 (17.8)	4.3 (0.9)	7.9 (1.9)	6.9 (1.7)	5.5 (0.9)	3.0 (1.8)	1.2 (1.8)	0.5 (1.8)	25

Abbreviations: 1RM, 1-repetition maximum; FS_{VL20}, fixed sets 20% velocity loss threshold; LVP, load–velocity profile; PBT, traditional 1RM-percent-based training; VS_{VL20}, variable sets 20% velocity loss threshold.

reduction (ML and TL), which was not statistically different from the PBT session (ES = 0.00–0.34). By contrast, the significantly lower mechanical stress (sCTUT and sTUT) during FS_{VL20} compared with PBT was due to the completion of fewer repetitions (ES = 1.01). A potential limitation of reduced mechanical stress associated with the LVP and FS_{VL20} sessions is the reduced training stimulus required to maximize muscle hypertrophy.[26] For example, Pareja-Blanco et al[9] had 22 resistance-trained males perform 8 weeks (16 sessions, ranging from ~68 to ~85% 1RM) of the back squat exercise on a Smith machine (back squat 1RM: 106.2 [13.0] kg; 1RM to body mass ratio: 1.41 [0.19]) with the training groups only differing by the allowed velocity loss threshold in each set (20% vs 40%). The 40% velocity loss training group performed significantly more repetitions (total repetitions: 310.5 [42.0] vs 185.9 [22.2]; mean repetitions/set: 6.5 [0.9] vs 3.9 [0.5]) and had significantly greater hypertrophy of vastus lateralis and intermedius muscles (9.0% increase in muscle volume) compared with the 20% velocity loss group (3.4% increase), which was not surprising because they completed significantly more work by performing 40% more repetitions.[26] These findings indicate that completing fewer repetitions or reducing mechanical stress likely results in less muscular development, suggesting that the FS_{VL20} method and parameters employed in the present study may not be beneficial for hypertrophic-oriented training. However, Pareja-Blanco et al[9] also found that the 20% velocity loss training group maintained significantly faster movement velocities (MV: 0.69 [0.02] $m \cdot s^{-1}$ vs 0.58 [0.03] $m \cdot s^{-1}$), had similar increases in maximal strength (18.0% vs 13.4%), and had significantly greater improvements in ountermovement jump height (9.5% vs 3.5%) compared with the 40% velocity loss group. Therefore, although more repetitions maximized muscle hypertrophy, more repetitions did not lead to additional strength gains and may not be advantageous for adaptations associated with explosive, powerful movements.

Although the present study did not investigate the chronic effects of these protocols, it is possible to hypothesize the rationale for adopting the LVP, FS_{VL20}, and VS_{VL20} training sessions for strength and power development in a strength-oriented training cycle. The phosphagen system in the human body is the predominant energy system responsible for the explosive movements desired to maximize increases in strength and power output.[1,27] This energy system typically lasts for up to 10 seconds of maximal effort and, when depleted, coincides with rapid decreases in movement velocity.[27] If energy stores are depleted without sufficient recovery, it is speculated that training under energy depletion and excessive velocity loss could induce adaptations toward slower, more fatigue-resistant fiber types. This is particularly important for athletes whose training goal is primarily focused on explosive force production associated with strength and power training and not on maximizing muscle hypertrophy. In addition, increased muscular development could be problematic for athletes required to maintain a specific body mass, and furthermore, greater mechanical stress, which does not lead to greater increases in strength, may cause unnecessary fatigue and prolong recovery time. Therefore, to optimize strength and power development, a coach can employ PBT and prescribe a lower number of repetitions per set, or VBT (eg, LVP, FS_{VL20}, and VS_{VL20}) with objective repetition velocity feedback to reduce the amount of velocity loss in a training session so that the required energy system and preferential targeting of type IIX fibers can be utilized to maximize strength and power development.[9,12]

Importantly, participants completed the VS_{VL20} session in a significantly shorter amount of time (~90 s shorter) compared with the other sessions, yet there were no significant differences in MV, PV, MF, PF, MP, PP, ML, TL, MW, TW, ETUT, CTUT, or TUT between VS_{VL20} and all other sessions. This additional 90 seconds per exercise (eg, ~9 min for 6 exercises) could potentially be reallocated to another training modality or an additional exercise: something that could be valuable during time-restricted strength sessions. Although some may argue that it takes more time to implement VBT compared with PBT due to setting up the devices, these steps can be done by strength and conditioning staff before training, which does not increase an athlete's training time. However, in the present study, there were 2 participants who took longer (6 sets) than the 5 sets prescribed in the other experimental sessions. Although VS_{VL20} required participants to complete as many quality (highest possible velocity output against a given load) repetitions in as few sets as possible, the VS_{VL20} training method can allow for flexibility in determining the optimal repetition scheme in order to accommodate daily fluctuations in performance.[28] As such, the VS_{VL20} method allowed these 2 participants to complete fewer repetitions per set, allowing for more recovery time in order to complete the prescribed total number of repetitions with higher velocity outputs. By contrast, VS_{VL20} also allowed the other 13 participants to complete 25 repetitions in fewer sets than the other experimental methods. Therefore, VS_{VL20} could be preferred over PBT because it integrates a more objective, individualized approach based on an athlete's readiness to train.

The inclusion of VBT methods over PBT may raise some potential limitations with its use in large groups. For example, all VBT methods require specialized equipment to accurately monitor movement velocity, and an athlete or coach must modify the training load or repetition volume based on the velocity outputs. Additionally, the LVP method requires individualized mathematical calculations, but these are no more difficult than the calculations required for PBT load prescription. Nevertheless, monitoring devices are becoming more affordable and the latest devices now provide instantaneous feedback, making it simple to employ the VBT methods from the present study. For instance, velocity-monitoring tools can report the average MV of a set, making it easy to compare with a prescribed target velocity (LVP method).

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Furthermore, some devices even allow for a specified velocity loss threshold to be set prior to training (eg, FS_{VL20} and VS_{VL20}), expediting their use during training.

Practical Applications

The present study shows that the VBT methods employed in the present study may serve as an alternative to more traditional strength-oriented PBT sessions. Specifically, the LVP and FS_{VL20} methods permitted individuals to perform repetitions with faster velocities across the entire training session compared with PBT, performing repetitions with less mechanical stress but maintaining similar measures of force and power output. Alternatively, the VS_{VL20} method had similar kinetic and kinematic data compared with PBT and the other VBT methods but could be completed in a significantly shorter time period, which could benefit individuals with time constraints. However, it must also be acknowledged that the use of VBT methods requires time to set up equipment prior to training, which is not required for PBT sessions.

Conclusions

The present study revealed that individuals employing the LVP and FS_{VL20} VBT methods could reduce mechanical stress and maintain significantly faster movement velocities during a training session compared with PBT. In addition, VS_{VL20} elicited similar training responses to the other experimental sessions, yet was completed in a significantly shorter time. Therefore, VS_{VL20} could be viewed as a viable training method for athletes who are pressured for time. As a consequence, the use of VBT allows one to modify training, accounting for the current state of the neuromuscular system. Results from the present study show that the LVP and FS_{VL20} VBT methods can be employed in a strength-oriented training phase to diminish fatigue-induced decreases in movement velocity that can occur in training based on 1RM percentages.

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